

Q1: Reliable Data Transfer

Protocol State machines for S and R1/R2

- sequence

S

S “sending” #: $\text{udt_send}(\text{pkt}) \rightarrow$ S “listening for ACK #”

S “listening for ACK”:

If

- $\text{udt_receive}(\text{pkt}) \ \&\& \ !\text{is_ack}(\text{pkt}, \#)$
- $\parallel \text{udt_receive}(\text{pkt}) \ \&\& \ \text{is_corrupt}(\text{pkt})$
- $\parallel \text{timeout}$

$\rightarrow$ S “sending #”

If $\text{udt_receive}(\text{pkt}) \ \&\& \ \text{is_ack}(\text{pkt}, 1/2) \ \&\& \ !\text{is_corrupt}(\text{pkt})$...

... If already_got(2/1) $\rightarrow$ S “sending # + 1” Else set_got(2/1)

R (1/2)

Same as S except it sends ACKs instead of messages isn't listening for 2 things.

Q2: Throttling

Flow Control: A procedure that helps prevent hosts from overwhelming each other. In TCP, each host will include the size of 'rwnd' (receive window – space in its receive buffer), and the responding host will not send more data than that size. When the rwnd is '0', the sender will periodically send 1 byte messages, waiting for the rwnd to increase.

Congestion Control: Procedures that manage congestion on a busy network. In TCP, we limit our sending limited by the 'cwnd' (congestion window), which is based on an estimate of network traffic. Traffic will be sent based on what state the system is in.

1) Slow Start: sends messages with a small cwnd that grows exponentially. We start here. When growth hits the ssthresh (slow start threshold) we switch to congestion avoidance. If we hit 3 duplicate ACKs we switch to Fast Recovery.

2) Congestion Avoidance: We increase cwnd more slowly until a timeout (back to Slow Start), or 3 duplicate ACKs (to Fast Recovery).

3) Fast Recovery: Maintains sending rate. When a new ACK is received we switch to Congestion Avoidance. After a timeout, we go back to slow start.

A more complete explanation would go into what cwnd and ssthresh are set to on transitions and receiving ACKs.

Q3: Routers

Subnet prefix: 1.1.0.0/21

Router A's forwarding table:

Router A	Subnet routing
Port 1	1.1.1.0/24
Port 2	1.1.2.0/24
Port 3	1.1.3.0/24
Port 4	Everything else.

Q4: Routing

